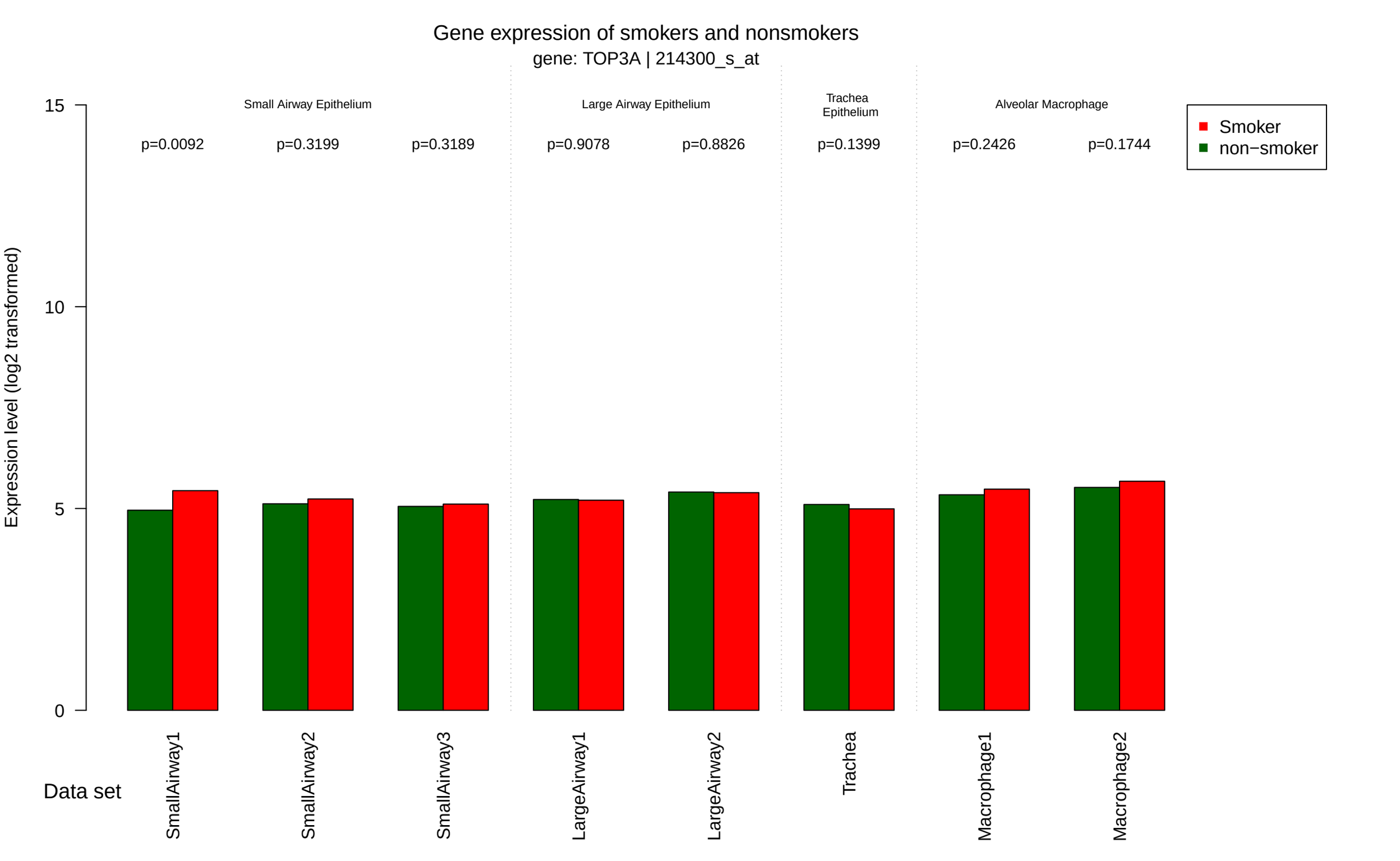
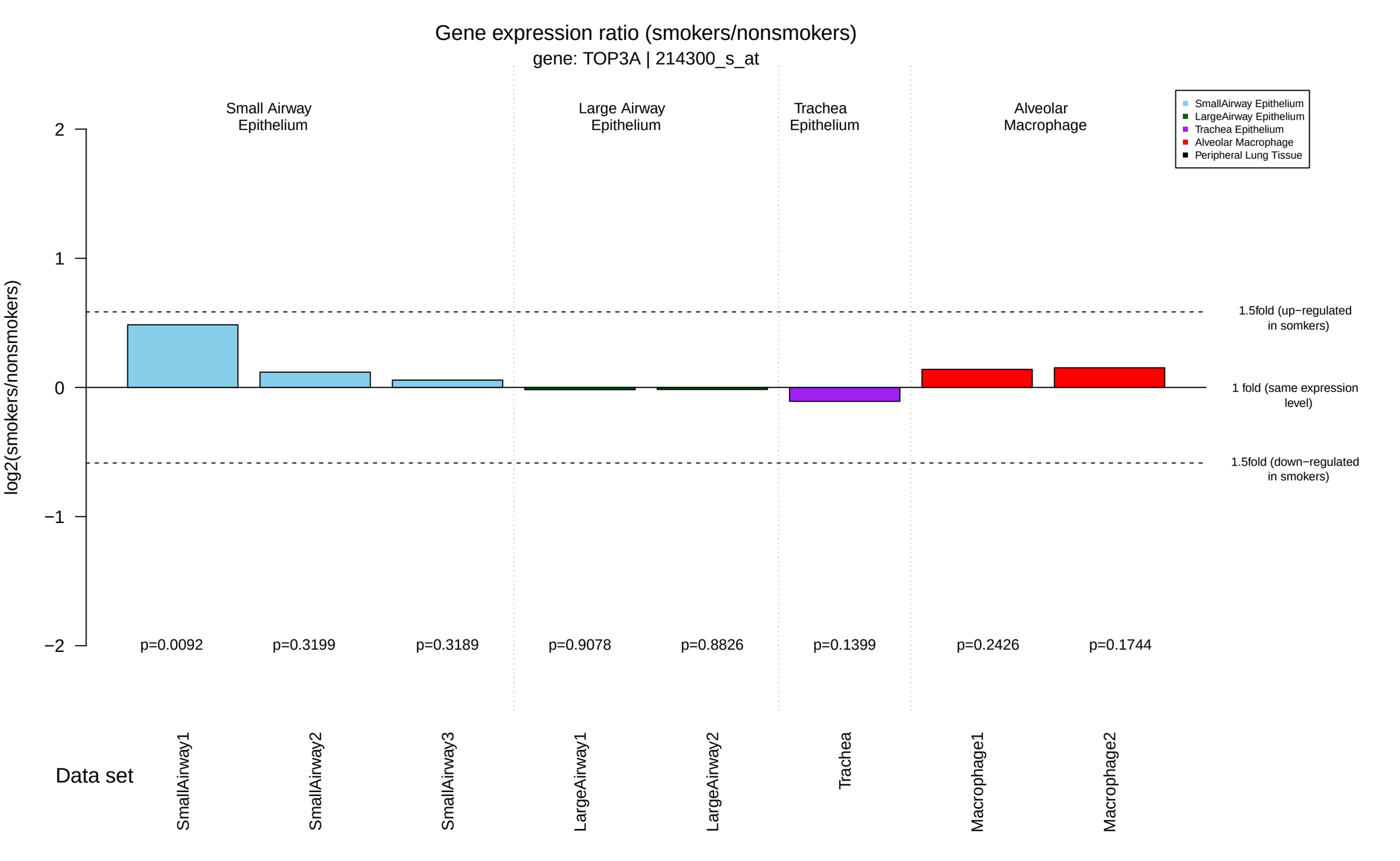
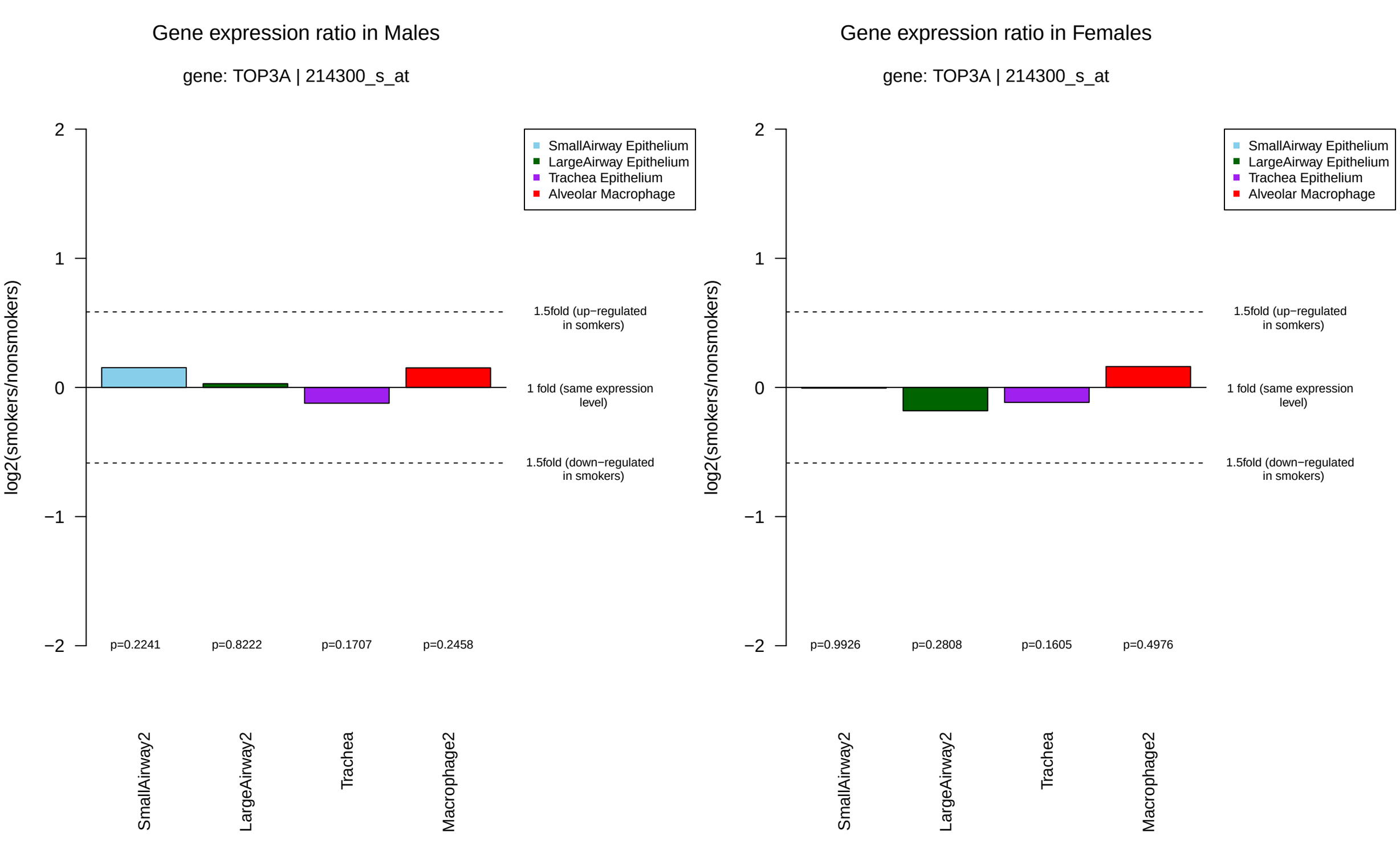
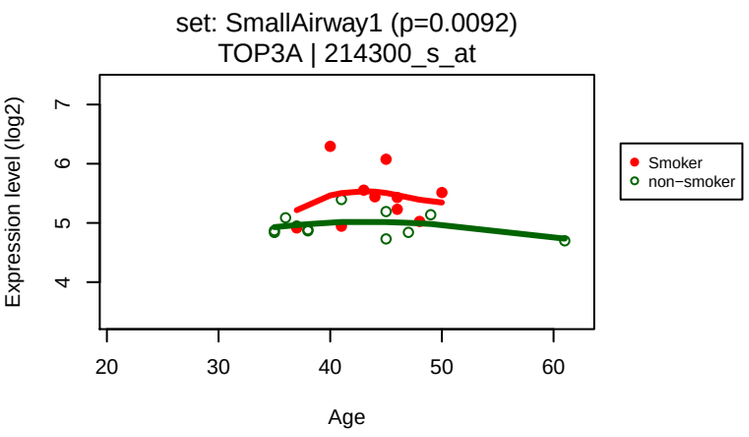
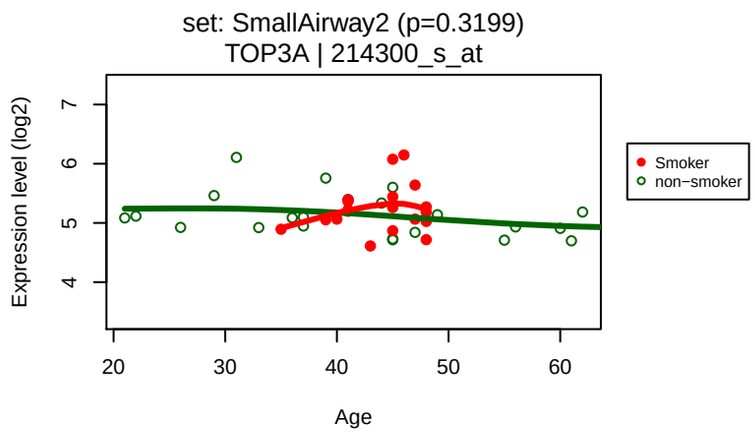
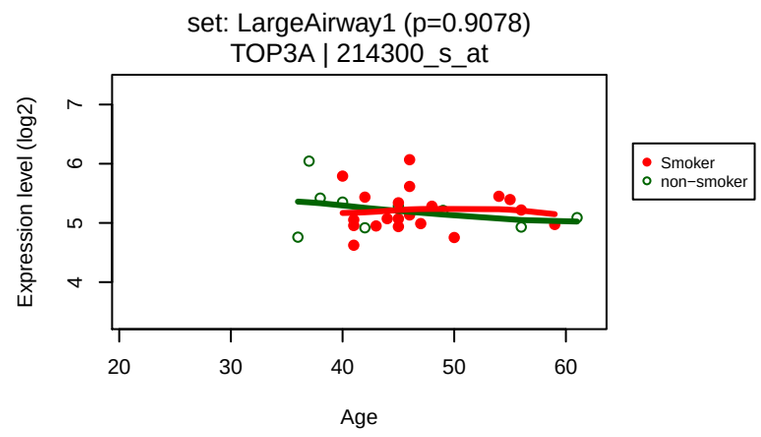

Gene expression of smokers and nonsmokers
gene: TOP3A | 214300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0092
p=0.3199
p=0.3189
p=0.9078
p=0.8826
p=0.1399
p=0.2426
p=0.1744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

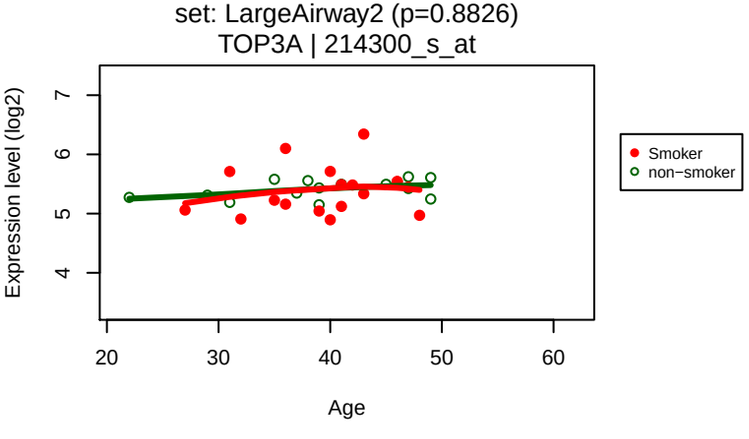

Gene expression ratio (smokers/nonsmokers)
gene: TOP3A | 214300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0092
p=0.3199
p=0.3189
p=0.9078
p=0.8826
p=0.1399
p=0.2426
p=0.1744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

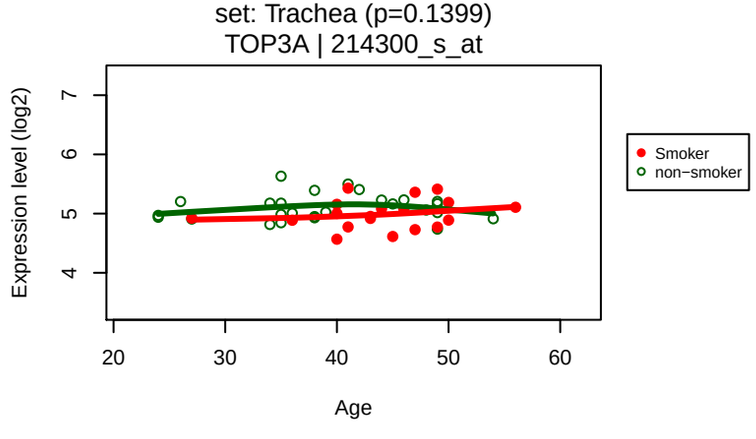

Gene expression ratio in Males
gene: TOP3A | 214300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2241
p=0.8222
p=0.1707
p=0.2458
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP3A | 214300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9926
p=0.2808
p=0.1605
p=0.4976
SmallAirway2
LargeAirway2
Trachea
Macrophage2

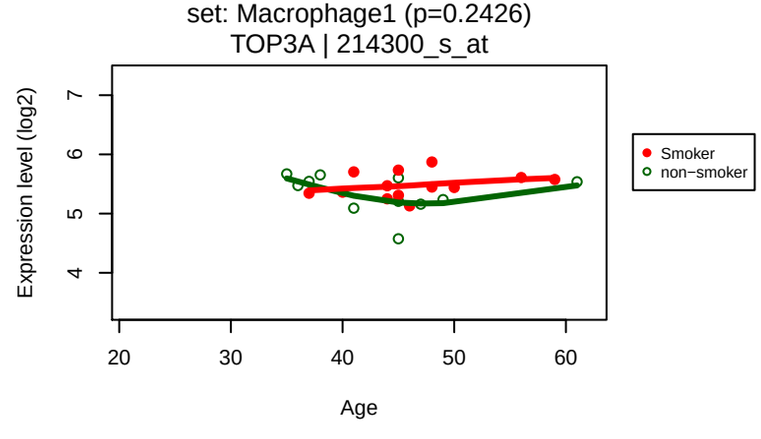

set: SmallAirway1 (p=0.0092)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker

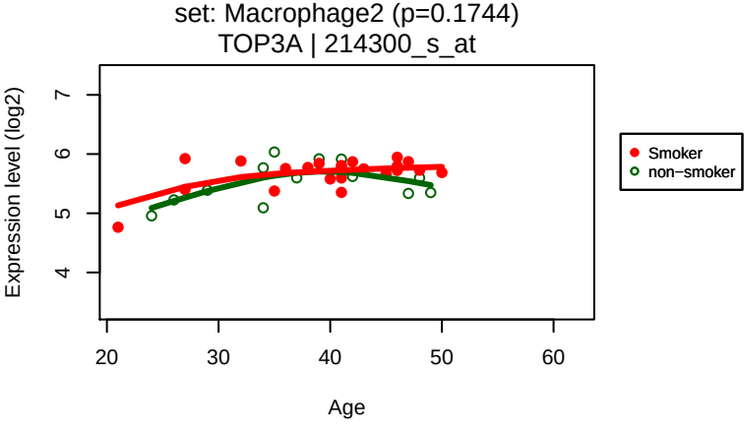

set: SmallAirway2 (p=0.3199)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker

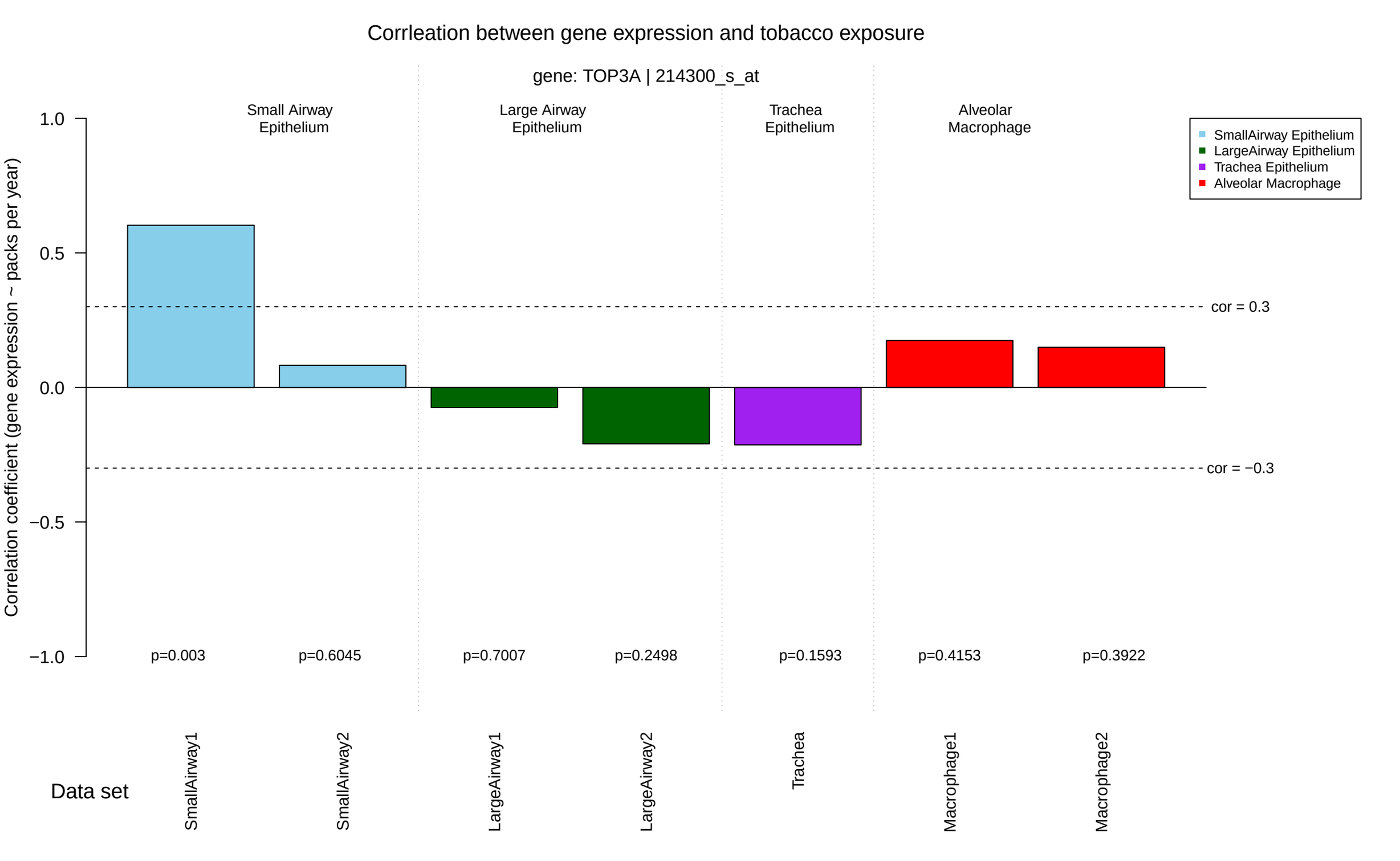

set: LargeAirway1 (p=0.9078)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8826)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1399)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2426)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1744)
TOP3A | 214300_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP3A | 214300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.003
p=0.6045
p=0.7007
p=0.2498
p=0.1593
p=0.4153
p=0.3922
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2